

3 Lab 2: Lap Time Simulation Pt.1

3.1 Introduction

The primary objective of this laboratory is to understand vehicle lap time sensitivity to various design parameters through simulation. Utilizing lap time simulations allows for the isolation of different physical effects, rapid setup iterations, and strategy development without the need for physical track testing.

The vehicle analyzed in this study is the **Tatuus FA010**. Its main specifications are summarized in the table below:

Parameter	Value
Height	950 mm
Axle track	1490 mm (front), 1455 mm (rear)
Wheelbase	2650 mm
Power	134 kW
Transmission	6-speed semi-automatic sequential gearbox

Table 12: Tatuus FA010 Specifications

The simulations are conducted across three distinct race tracks: **Spa-Francorchamps, Monza, and Donington**.

To perform these analyses, we utilize OpenLap, an open-source MATLAB code. The software suite consists of four modules:

- **OpenLap** for point-mass quasi-static simulations
- **OpenVehicle** for generating vehicle models
- **OpenTrack** for track modeling
- **OpenDrag** for straight-line acceleration and braking simulations

3.2 LapTime Simulation: Quasi steady State vs Transient

When simulating a vehicle's performance over a lap, two primary approaches can be adopted: transient and quasi-steady-state simulations.

In a **transient simulation**, the vehicle is modeled as continuously experiencing transient conditions, meaning dynamic weight transfer, inertia, and mass effects are fully considered as the car runs along the track.

Conversely, the **quasi-steady-state simulation**, adopted in this laboratory, does not consider transitions. Instead, it calculates the vehicle's maximum performance capabilities at discrete points along the track.

3.2.1 Quasi Steady State Simulation: Code Logic

The OpenLap quasi-steady-state simulation determines the ultimate lap time through a sequential five-step process:

1. **Maximum speed in each sector:** Given a fixed discretized trajectory, the simulation calculates the maximum sustainable speed for mid-corner conditions (where longitudinal acceleration $a_x = 0$).

$$v = \sqrt{ggv(v, a_x = 0) * R}$$

This is found iteratively using the vehicle's Performance Envelope (GGV map), which dictates the maximum available lateral acceleration.

2. **Determination of minimum speed points:** The solver identifies the mid-corner points where the vehicle's grip is fully saturated by lateral acceleration. These points correspond to the local minima of the speed profile calculated in the first step.
3. **Corner entry simulation:** To evaluate the braking profile, the simulation performs a **back-integration** starting from the corner apex (the local speed minima). It uses the GGV diagram to extract the available longitudinal deceleration (a_x) that saturates the combined grip, assuming uniformly accelerated motion to compute the entry speed profile using the equation:

$$\text{At apex } i (v_i, s_i): v_{i-1} = \sqrt{v_i^2 + 2 \cdot a_{x(i-1)} \cdot (s_i - s_{i-1})}$$

$$\text{where } a_{x(i-1)} = GGV(v_i, a_{y,(i-1)})$$

4. **Corner exit simulation:** Similar to corner entry, the exit profile is calculated by **integrating forward** from the apex. The simulation evaluates the available longitudinal acceleration (a_x) at the exit and computes the resulting traction speed profile.

$$\text{At apex } i (v_i, s_i): v_{i+1} = \sqrt{v_i^2 + 2 \cdot a_{x(i+1)} \cdot (s_{i+1} - s_i)}$$

$$\text{where } a_{x(i+1)} = GGV(v_i, a_{y,(i+1)})$$

5. **Final speed profile generation:** The ultimate speed profile for the entire lap is determined by taking the minimum speed value among the three calculated constraints at any given point:

$$v = \min(v_{brake}, v_{traction}, v_{cornering})$$

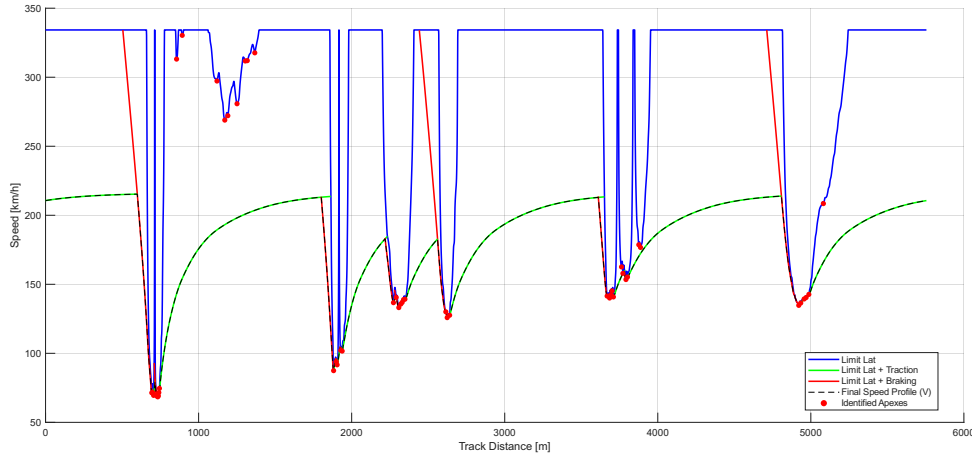


Figure 58: QSS Speed Profile Calculation on Monza

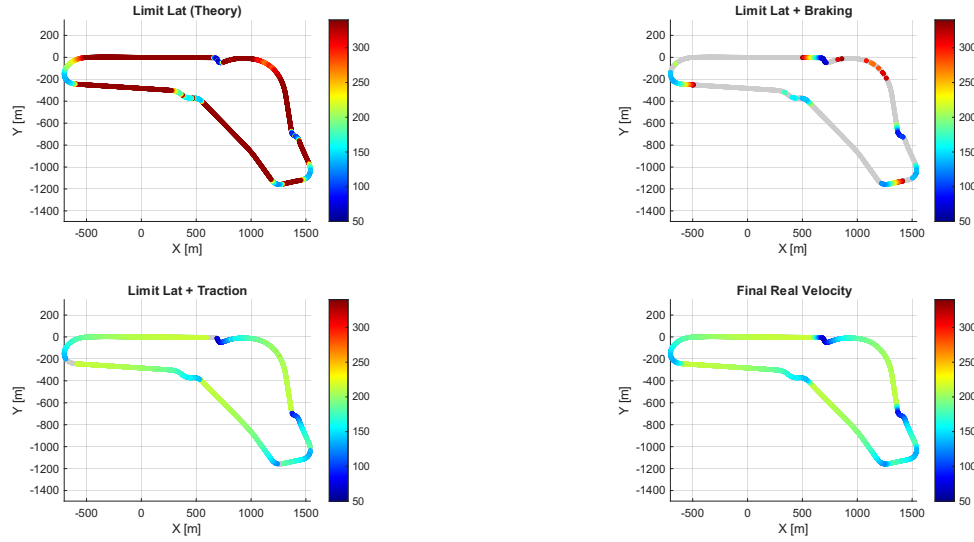


Figure 59: QSS Speed Profile Calculation Visualization on Monza Track

3.3 Track Fuel Effect Analysis

During a race stint, the progressive consumption of fuel reduces the total mass of the vehicle. A lighter car directly translates into higher longitudinal accelerations, shorter braking distances, and higher cornering speeds due to reduced inertial forces. To thoroughly evaluate this phenomenon, a two-step analytical methodology is implemented.

3.3.1 Stint Simulation and Linearity Validation

Initially, an iterative quasi-static simulation of a 10-lap stint was performed across the three different circuits (Monza, Spa-Francorchamps, and Donington). At the end of each lap, the exact amount of consumed fuel was calculated and subtracted from the vehicle's total mass for the subsequent iteration.

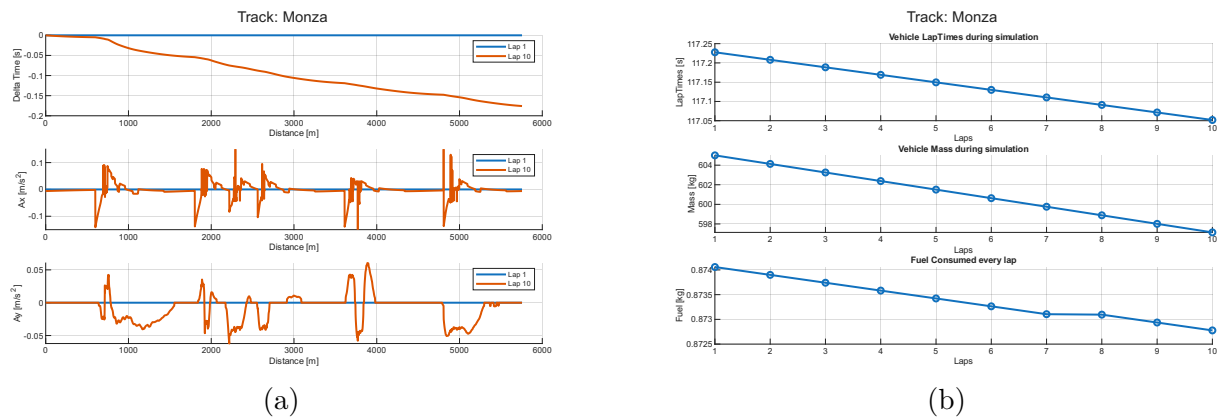


Figure 60: Monza

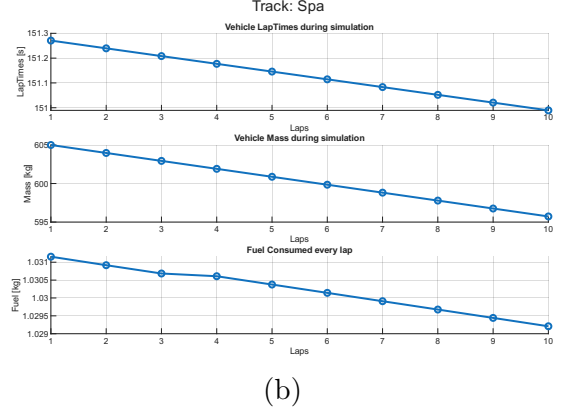
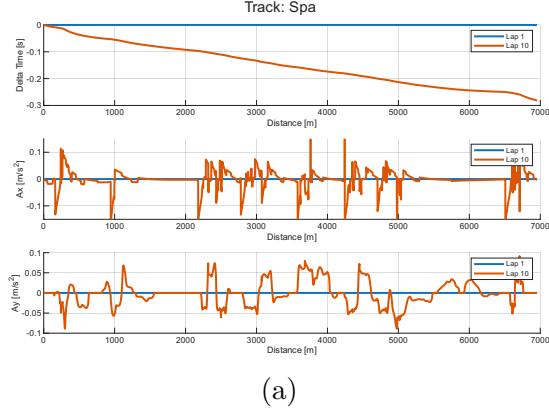


Figure 61: Spa-Francorchamps

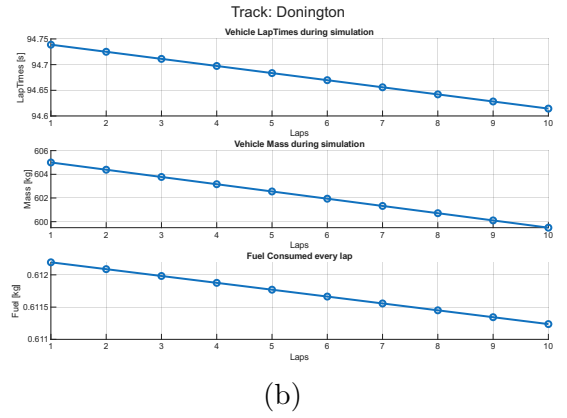
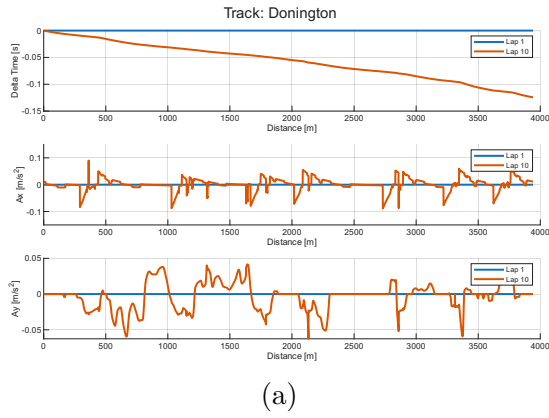


Figure 62: Donington

Looking at the telemetry Plots between the first and last laps, it's clear that the lighter vehicle in Lap 10 is overall faster. However, the largest amount of time gain is done in correspondence of traction and braking zones, where the inertia of the vehicle has a significant impact.

As shown in the lap time evolution graphs, the performance gain follows a quasi-linear trend. Since the fuel burned over 10 laps represents a minimal fraction of the total vehicle mass, the complex non-linear dynamics of tire load sensitivity and inertial equations can be safely linearized. Therefore, the local derivative of lap time with respect to vehicle mass can be confidently approximated as constant

3.3.2 Normalized Index Evaluation

Having validated the linear behavior of the vehicle within this mass window, a precise fuel load sensitivity index is calculated. Two independent simulations are executed for each track:

- **Baseline Configuration:** Baseline vehicle mass ($M_{heavy} = 605$ kg).
- **Light Configuration:** Vehicle mass artificially reduced by a discrete step of 10 kg ($M_{light} = 595$ kg).

By enforcing an exact 10 kg delta, a solid and reliable time difference (Δt_{lap}) is extracted. To provide a robust and track-independent engineering metric, a **Normalized Fuel Sensitivity** index (S_{fuel}) can be formulated. This index normalizes the absolute time gained by the total length of the track:

$$S_{fuel} = \frac{t_{heavy} - t_{light}}{L_{track}}$$

where t_{heavy} and t_{light} are the respective lap times in seconds, and L_{track} is the circuit length in kilometers. The resulting metric expresses the *seconds lost per 10 kg of fuel per kilometer of track* $\left[\frac{s}{10kg \cdot km} \right]$.

The following histogram compares this index across the analyzed tracks, highlighting how circuits with a higher frequency of stop-and-go sections exhibit a higher sensitivity to mass variations.

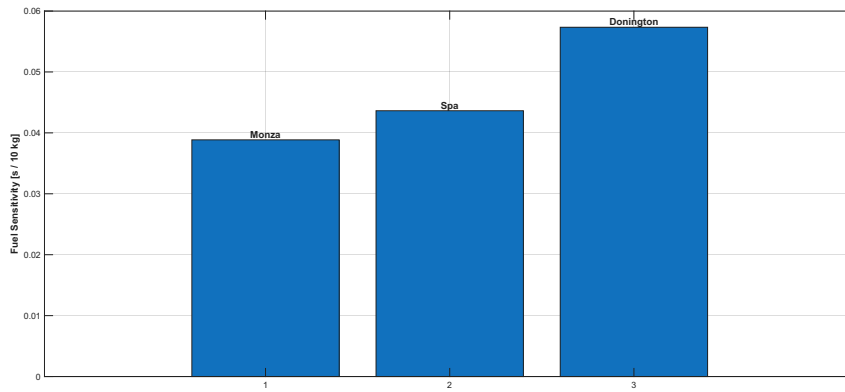


Figure 63: Normalized Fuel Sensitivity: Time lost per 10 kg of mass reduction, normalized by track length.

3.4 Aero setup choice

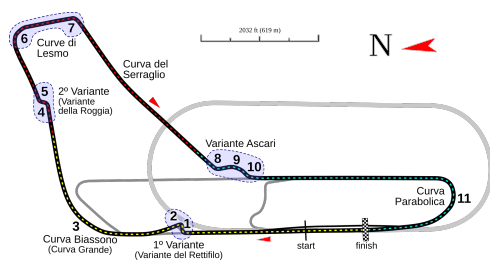
To optimize the vehicle's aerodynamic efficiency, we analyzed different aerodynamic setups. The available configurations for the Tatuus FA010 vehicle range between the following lift and drag coefficient boundaries:

Coefficient [-]	Range [-]
C_l	[0.5 - 2]
C_d	[0.6 - 1]

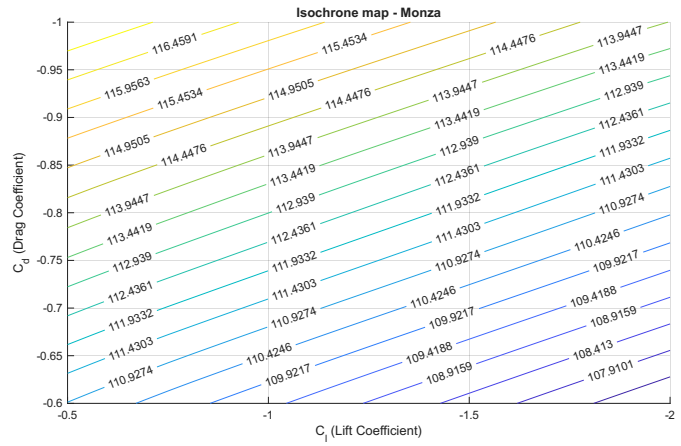
Table 13: Tatuus FA010 Aero Specifications

3.4.1 Isochrones Computation

By mapping the isochrone curves for respectively for Monza, Spa-Francorchamps and Donington, we can visualize how the trade-off between downforce and drag influences the overall lap time on each specific circuit geometry.

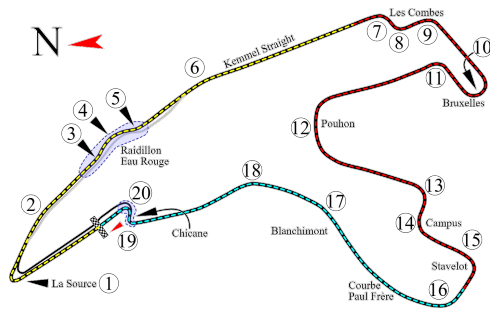


(a) Track Layout

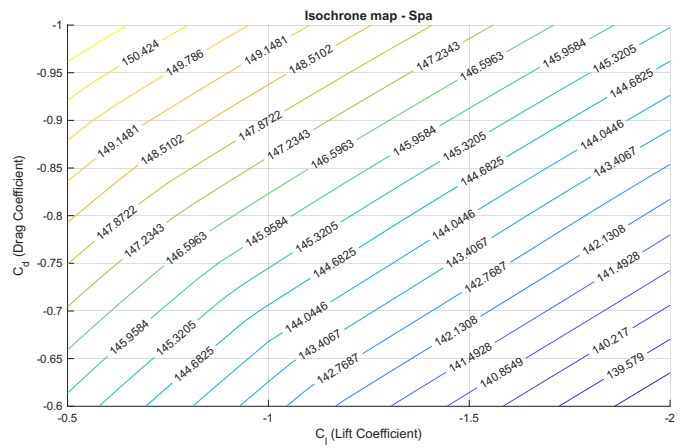


(b) Isochrones

Figure 64: Monza

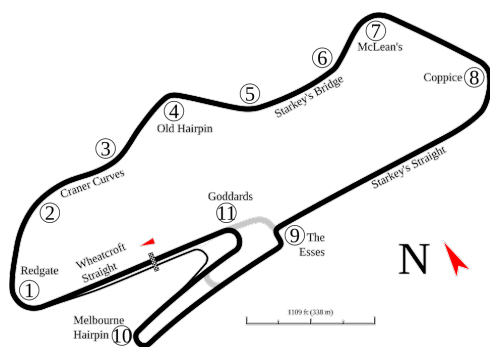


(a) Track Layout

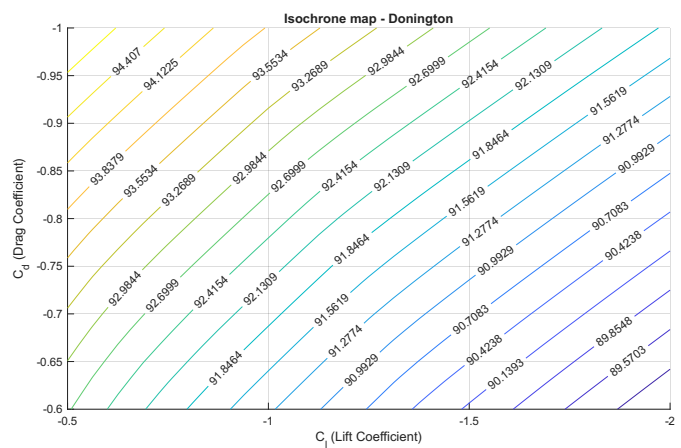


(b) Isochrones

Figure 65: Spa-Francorchamps



(a) Track Layout



(b) Isochrones

Figure 66: Donington

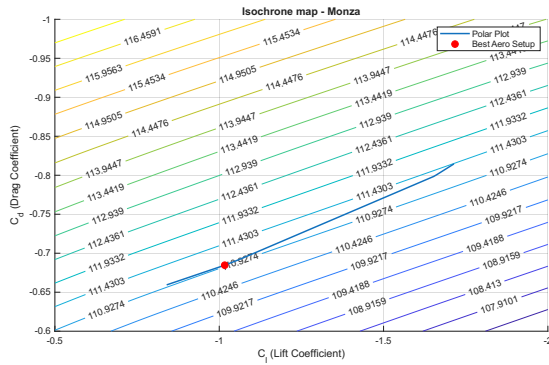
The isochrones plotted represent the specific ”**exchange rate**” of the circuit: they indicate how much drag can be added for a unit of downforce while maintaining the same lap time. The slope of these curves varies drastically depending on the track layout:

- **Slow circuits (e.g., Donington):** The isochrone is **steep**. The lower average speed makes drag less penalizing, allowing aerodynamic efficiency to be sacrificed in order to maximize cornering downforce.
- **Fast circuits (e.g., Monza):** The isochrone **flattens**. Due to the high speeds on long straights, the power dissipated by drag ($P \propto v^3$) dominates the lap time. Even a minimal increase in drag is highly detrimental, making the circuit extremely sensitive to overall aerodynamic efficiency.

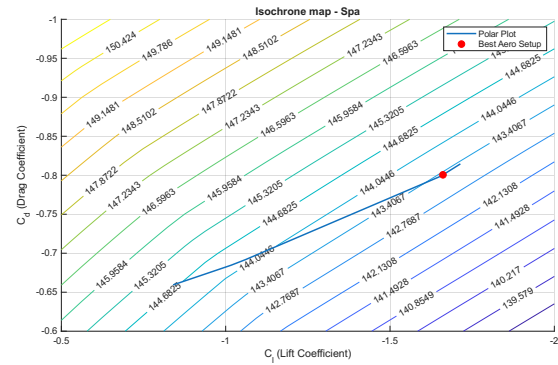
3.4.2 Polar Curve Plot and Final Setup Considerations

By overlaying the vehicle’s specific aerodynamic polar curve (the physical combinations of C_l and C_d the car can actually achieve) into the isochrone maps, we can identify the optimal aero setup.

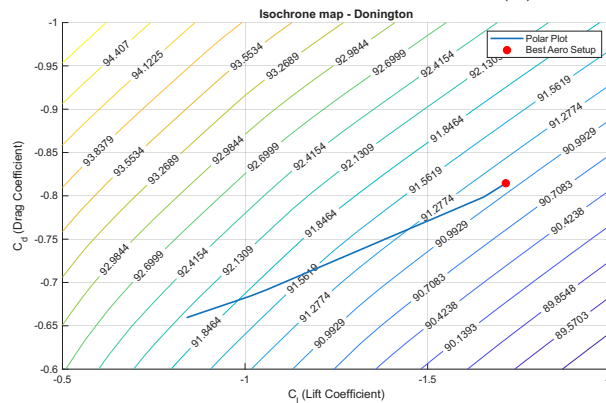
The best theoretical configuration is the point on the polar curve that acts as a tangent to the lowest possible lap time isochrone.



(a) Monza



(b) Spa-Francorchamps



(c) Donington

Circuit	C_L [-]	C_D [-]	Estimated Lap Time [s]
Monza	1.017	0.685	110.96
Spa	1.661	0.801	143.34
Donington	1.714	0.815	91.08

Table 14: Optimal Aerodynamic Setup

The optimal aerodynamic setups for Monza and Donington are aligned with the predictions: a low-downforce configuration is favored for the long straights of Monza, whereas a maximum-downforce package is necessary to maximize grip along the twisty-layout of Donington.

Spa-Francorchamps Counter-intuitive Optimal Setup: The simulation results for Spa-Francorchamps show a curious anomaly. Despite Spa being a famous fast circuit where a medium or low-drag setup is typically preferred, the lap time simulation identifies a high-downforce setup as the optimal choice.

To investigate this counter-intuitive behavior, a comparative telemetry analysis was performed on the Spa-Francorchamps circuit, testing three distinct aerodynamic configurations: Low, Medium, and High downforce.

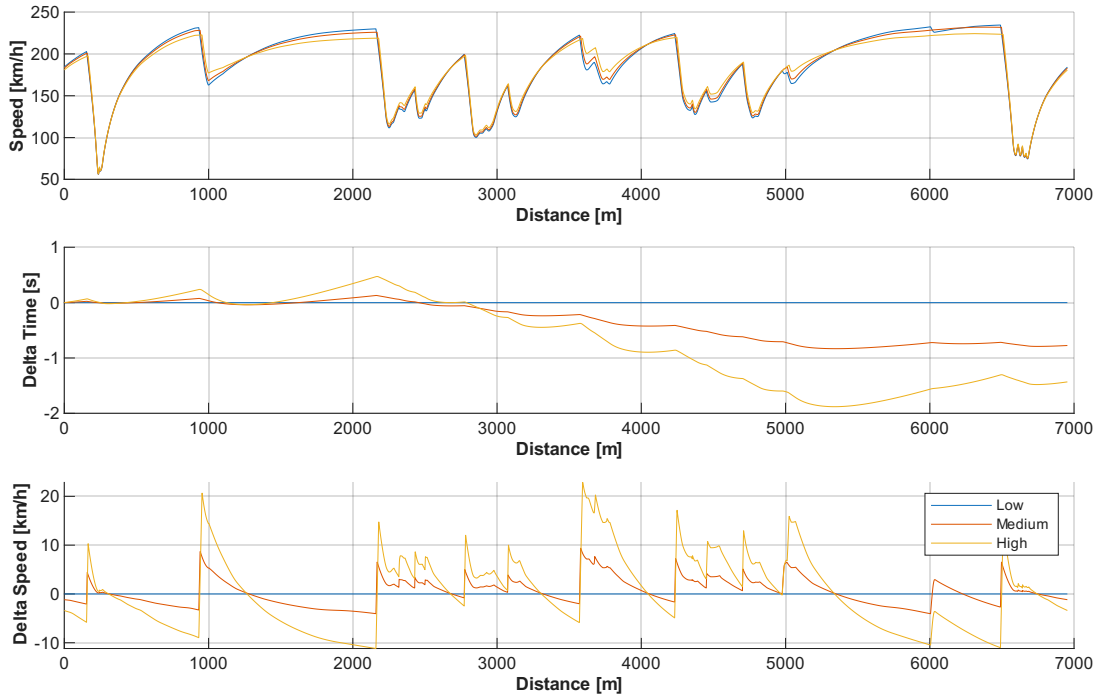


Figure 68: Telemetry

The Tatuus FA010 is an entry-level formula car equipped with a relatively modest PowerTrain delivering 134 kW. Due to this limited engine torque, the vehicle struggles to quickly recover momentum if the minimum apex speed drops too much. Any high deceleration (due to low-downforce setups) would cause severe drops in speed and engine RPMs, forcing the powertrain to slowly recover speed during the corner exit phase and costing a substantial amount of time.

On the contrary, the high-downforce configuration allows the vehicle to carry a much higher minimum speed through medium and fast corners. The telemetry unequivocally proves that the massive amount of time gained by preserving kinetic energy and momentum through corners completely overlaps the top-speed penalty induced by the aerodynamic drag on the straights.

3.4.3 Beyond Lap Time Simulation: Additional KPIs for Aero Setup

Relying only on the theoretical optimal lap time is not sufficient for real-world racing applications. When searching for the aerodynamic setup, race engineers must consider several other aspects that may influence the final setup choice:

- **Top Speed (Race-craft management):** A high-downforce setup generates more drag, significantly reducing the vehicle's top speed at the end of straights. In a race scenario, a car with low top speed is more vulnerable to being overtaken and will struggle to pass competitors.
- **Tire Management and Degradation:** Aerodynamic downforce pushes the car into the track surface, increasing vertical load and reducing sliding during cornering. Less sliding reduces thermal degradation. Therefore, a higher downforce setup is often preferred for race stints to extend tire life.
- **Fuel Consumption:** Higher aerodynamic drag requires the engine to spend more energy to maintain high speeds, leading to increased fuel consumption. This may force the car to start the race with a heavier fuel load or require the driver to manage fuel through 'lift and coast' techniques.
- **Cooling:** Due to external circumstances like external air temperature or quality, aerodynamics can be compromised if the car needs to meet the cooling demands of internal components (engine, batteries, etc...). This may go against pure aero performance to safeguard the reliability and efficiency of vital subsystems.